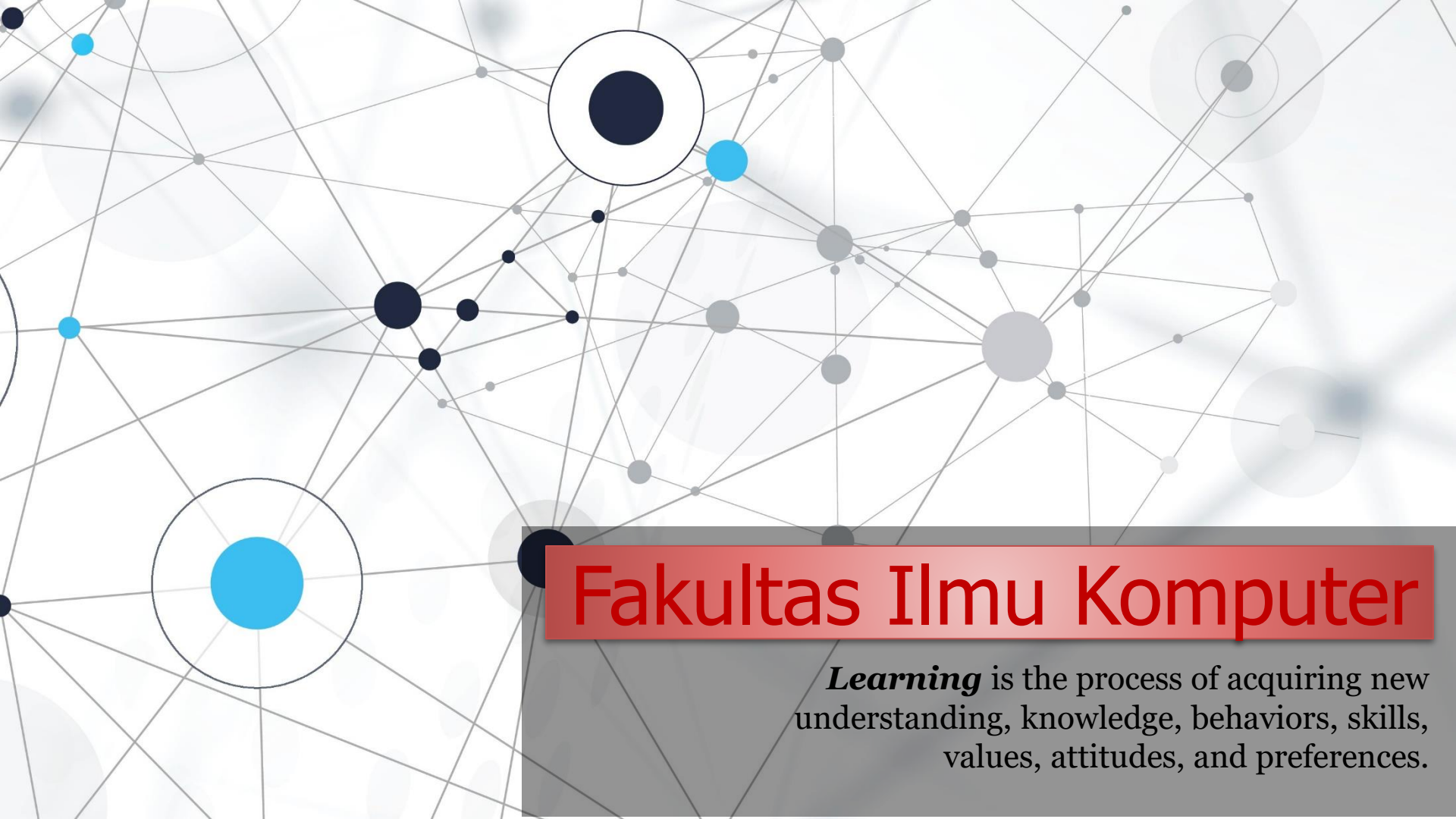


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

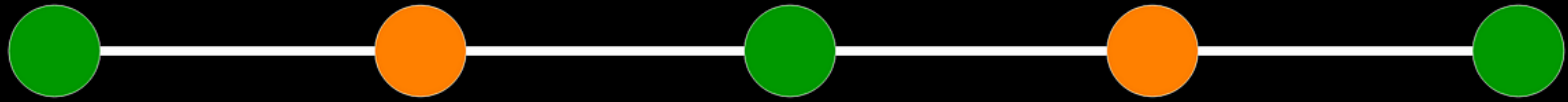
OOP in Python – Data Structures



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1 - Data Structures

A decorative border of green leaves and branches at the bottom of the slide.



DATA STRUCTURES



- ❖ A **data structure** is a **storage** that is used to **store** and **organize data**. It is a way of arranging data on a computer so that it can be accessed and updated efficiently.
- ❖ **What is a data structure?** Dalam istilah ilmu komputer, sebuah struktur data (*data structure*) adalah cara **penyimpanan**, **penyusunan** dan **pengaturan data** di dalam media penyimpanan komputer sehingga data tersebut dapat digunakan secara efisien.
- ❖ A **data structure** is not only used for organizing the data. It is also used for **processing**, **retrieving**, and **storing data**.
- ❖ There are different basic and advanced types of data structures that are used in almost every program or software system that has been developed.

Popular types of Data Structures:

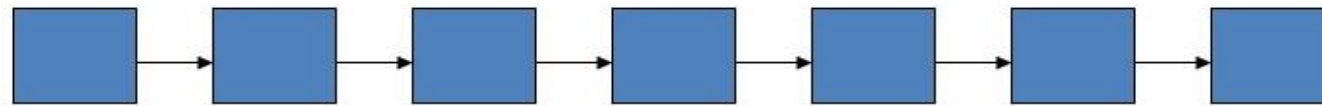
- Array
- Linked List
- Stack
- Queue
- Binary Tree
- Binary Search Tree
- Heap
- Hashing
- Graph
- Matrix
- Misc
- Advanced Data Structure

What is Data Structure?

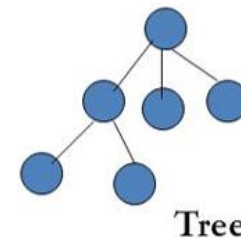
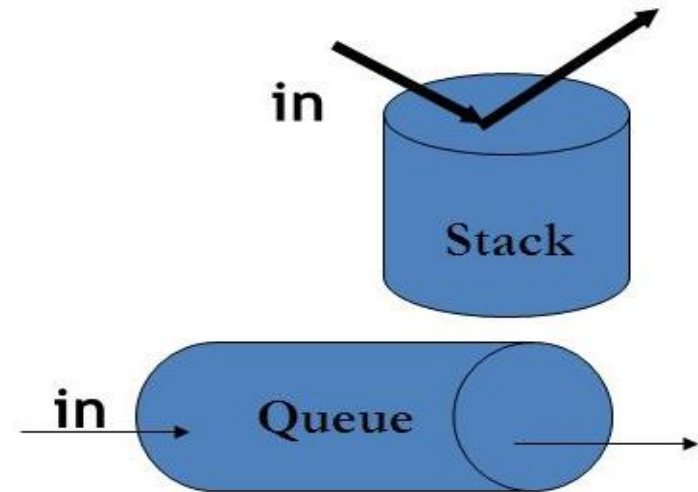
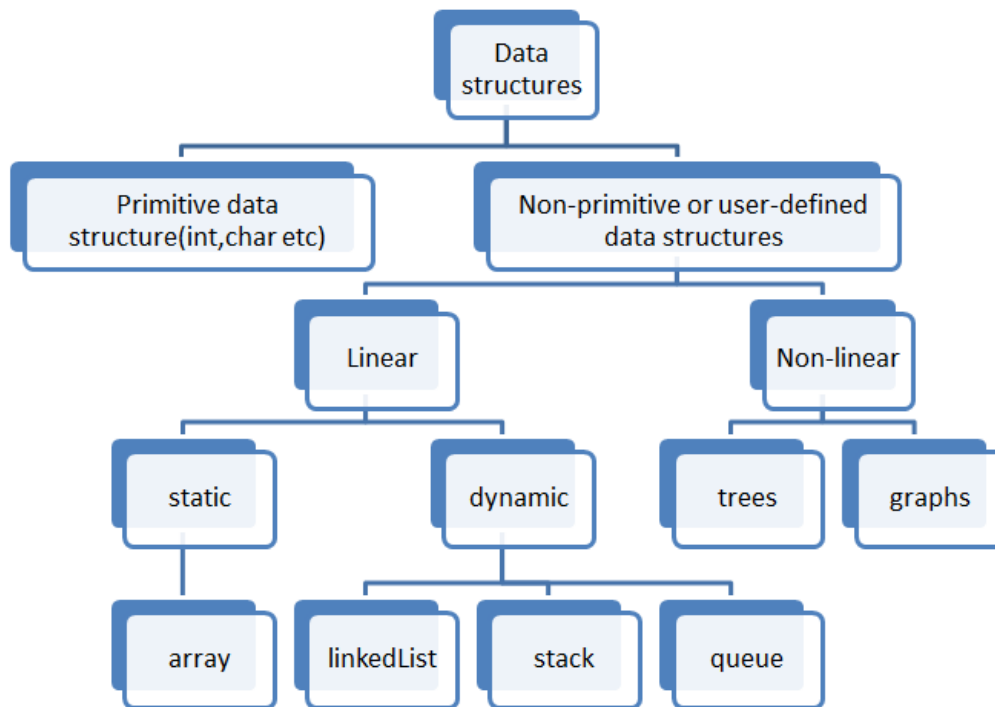
Data may be organized in different ways. We can represent the data in a particular way, so that it can be used efficiently.



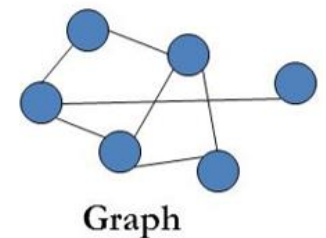
Array



Linked List



Tree



Graph

What is a Data Structure?



- ❖ **Data structure** is a collection of data that has specific ways of **accessing**, **storing**, and **organizing** the data. A data structure also defines any relationship between multiple pieces of data. It's like a **container**.
- ❖ A data structure (**container**) that has a ordered series of numbered slots that can each contain a piece of data. We access a piece of data with an index. Usually stores homogeneous data.

40	55	63	17	22	68	89	97	89
0	1	2	3	4	5	6	7	8

<- Array Indices

Array Length = 9

First Index = 0

Last Index = 8

Java Data Structures

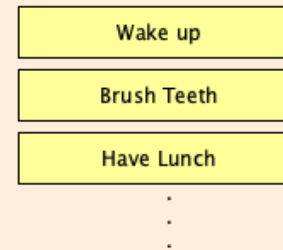
Array

Ordered, fixed-size, indexed by position, elements packed together in memory



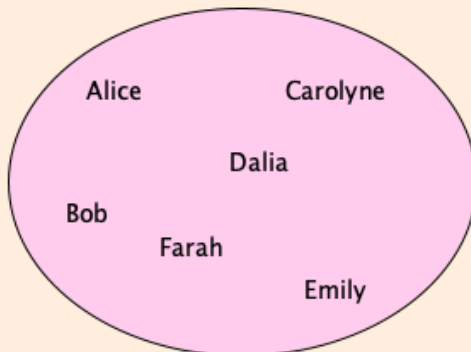
List

Ordered, variable size, comes in two main variants: ArrayList and LinkedList



Set

Unordered, no duplicates

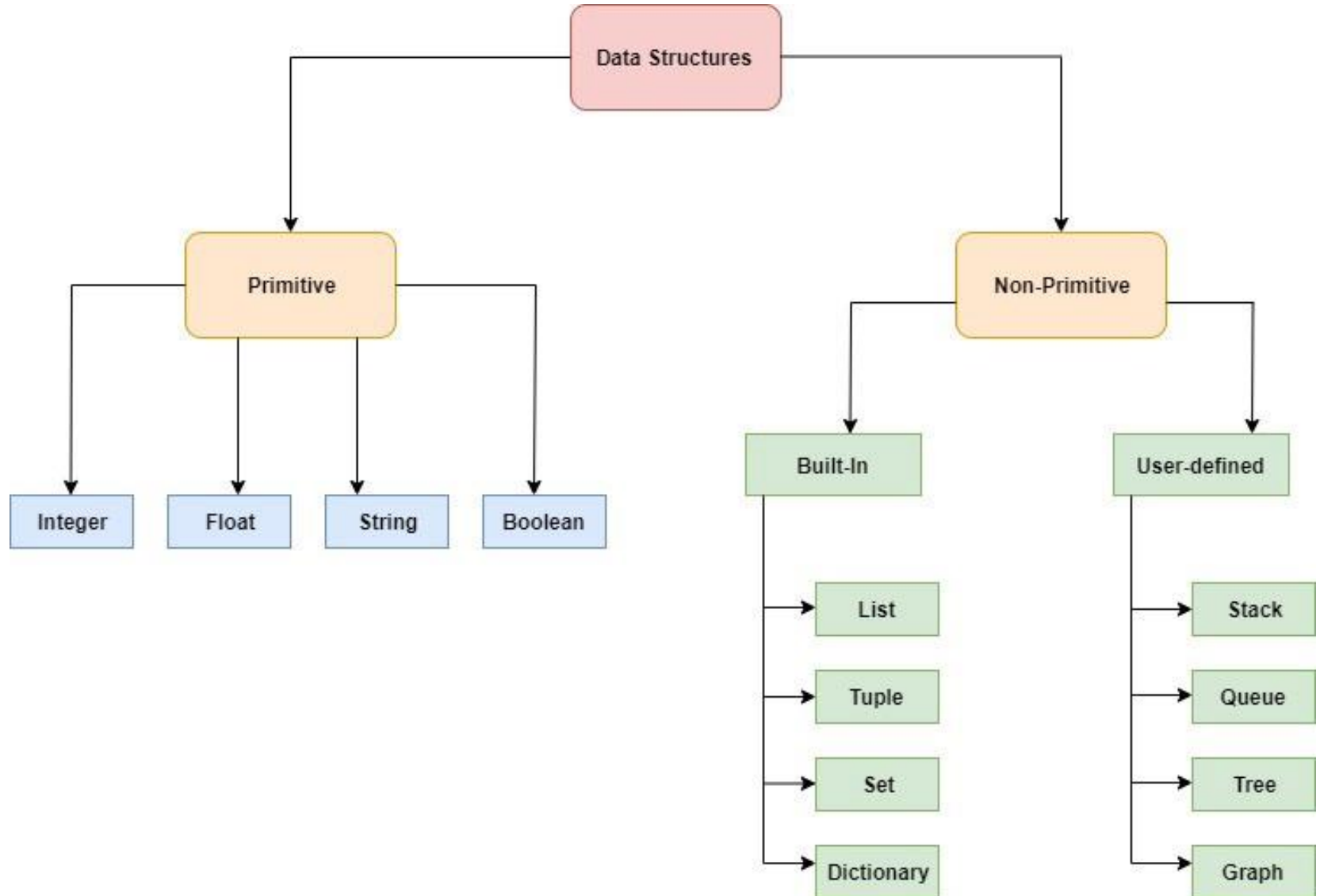


Map (Dictionary)

Unordered, key-value pairs, all keys unique

Alice	32
Carolyn	398
Bob	18
Dalia	27
Emily	809
Farah	11.29

Python Data Structures





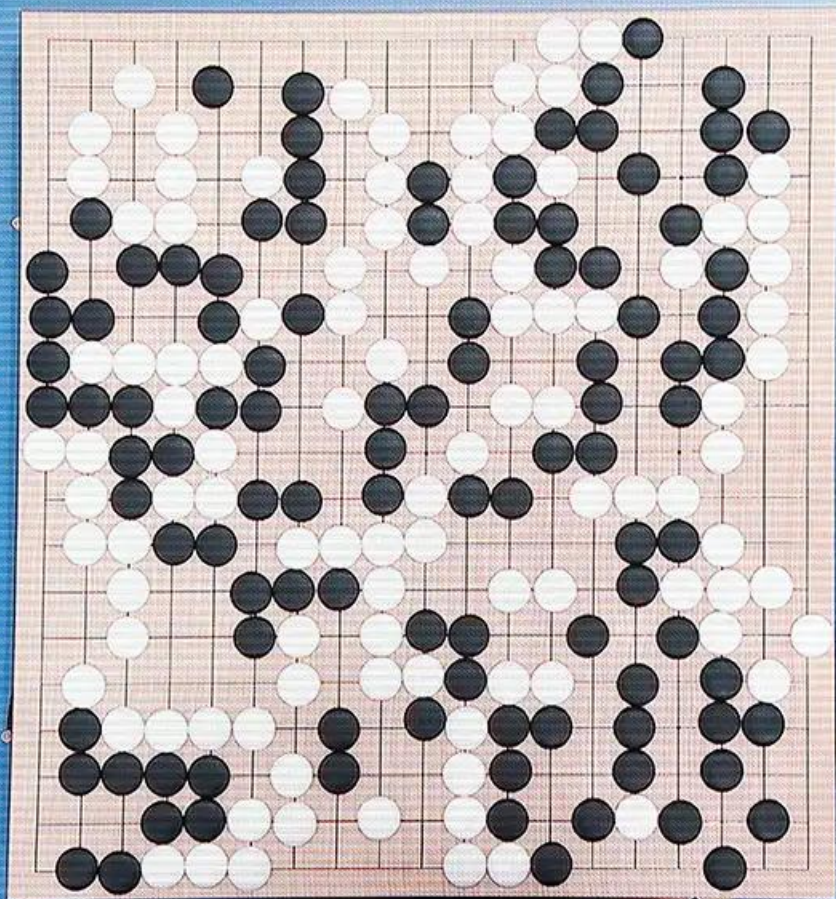
Data structure is the way data is arranged, stored, and organized in a computer program.

*Many security algorithms today, seperti algoritma enkripsi, hashing, atau autentikasi, melibatkan penggunaan **struktur data khusus** untuk mengelola dan memproses data dengan cara yang aman dan efisien.*



Data structure is used to represent and store data used in training and testing models. Efficient data representation can affect the **performance and speed of machine learning algorithms**.

An efficient data structure enables **fast access and manipulation of data**, which is essential in processing **large datasets**. Machine learning and deep learning algorithms require fast access to optimize model training and inference time.



柯洁 KE JIE
00:17:05



ALPHAGO
01:51:38

浙江省体育局



AlphaGo is an AI program developed by DeepMind (March 2016), using data structures as an integral part of its design and implementation. AlphaGo uses multiple neural networks to make decisions and predictions in the game of Go. Data structures are used to store the weights and architecture of these neural networks.

A hand is holding a tablet that displays the ChatGPT interface. The background is dark with some blue and green light effects. The tablet screen shows the 'ChatGPT' title at the top, followed by three sections: 'Examples', 'Capabilities', and 'Limitations'.

ChatGPT

Examples

"Write a poem about the future of AI."

"List five ways to improve your productivity."

"How do I use ChatGPT to help me learn?"

Capabilities

Remembers what you said earlier in the conversation.

Allows you to provide follow-up corrections.

Trained to decline inappropriate requests.

Limitations

May occasionally generate incorrect information.

May occasionally produce harmful instructions or hate speech.

ChatGPT developed by OpenAI and announced by OpenAI in June 2020. GPT-3.5 which is the basis for ChatGPT is a model based on the GPT-3 (Generative Pre-trained Transformer version 3) architecture. ChatGPT uses various data structures as part of its implementation. The data structures are used to store and manage the information required in the context of the conversation, to keep track of the context, understand the questions, and generate appropriate responses.



Orang-orang hebat seperti Elon Musk, Mark Zuckerberg, Sam Altman, dan Sundar Pichai tidak hanya terampil dalam pemahaman konsep dasar seperti *data structure*, tetapi mereka juga memiliki tim ahli dalam pengembangan *smart system*. Penggunaan *data structure* dalam pengembangan *advanced* dan *smart system* seperti yang mereka lakukan dapat mencakup **“algoritma dan optimisasi”, pengolahan big data, manajemen memori, “optimisasi database”, “artificial intelligence and machine learning”** dan bahkan pada **“pengembangan framework dan library”**.

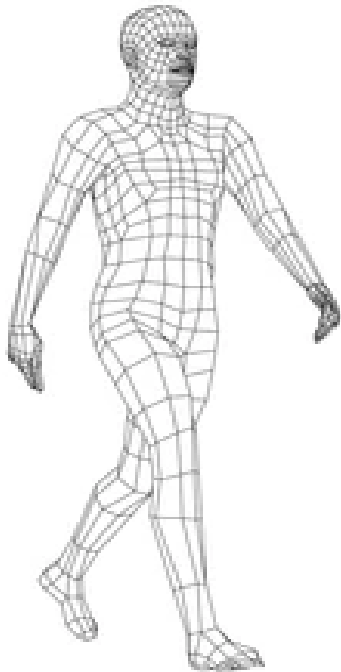
A decorative border of green leaves and branches at the top of the slide.

2 - What is data structure in OOP?

A decorative border of green leaves and branches at the bottom of the slide.

Data Structure *in* OOP

Object Oriented Programming (OOP) adalah tentang memperlakukan data sebagai objek (*objects*). Segala sesuatu dalam hidup dapat direpresentasikan sebagai objek dalam *code*. Namun, kita harus memiliki **blueprint (Class OOP)** sebelum kita membangun apapun.



Blueprint Person

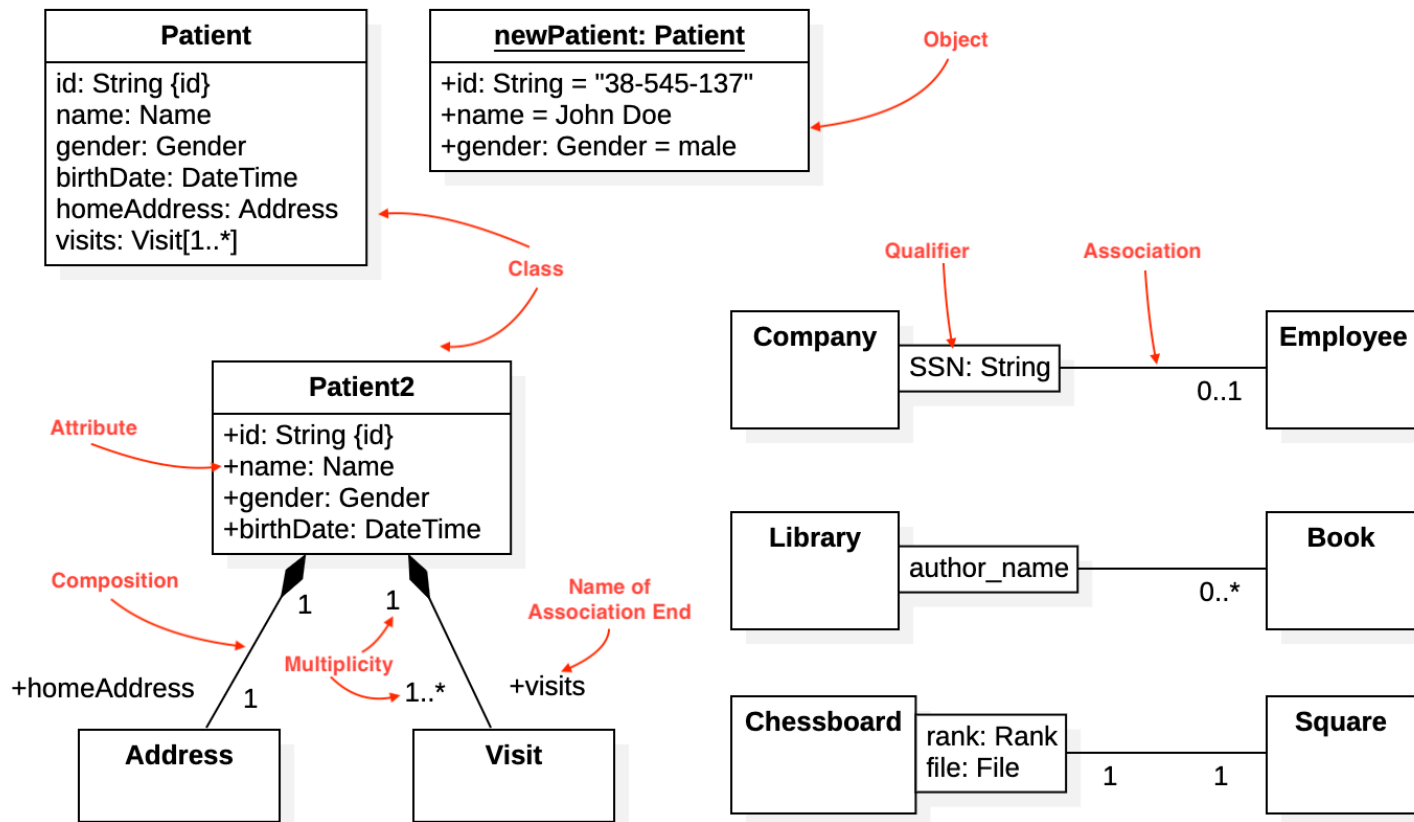
Blueprint (called a class in code): **Person**

- 1) String Name
- 2) Integer Age
- 3) String Eye_Color

- ❖ The **blueprint contains the structure that will hold the pieces of data**, we care about for a given person. We can create a person with something called a **constructor**, with a special type of function that will allow us to **create people** (*also called people instances*).
- ❖ A **blueprint** or **class** contains **properties** (name, age, eye color), a **constructor**, and **behavior** (*functions that access/alter the values of our properties or calculate related things*).
- ❖ These **behaviors** or **functions** are often called **methods** because they are tied/involved to the class/blueprint.

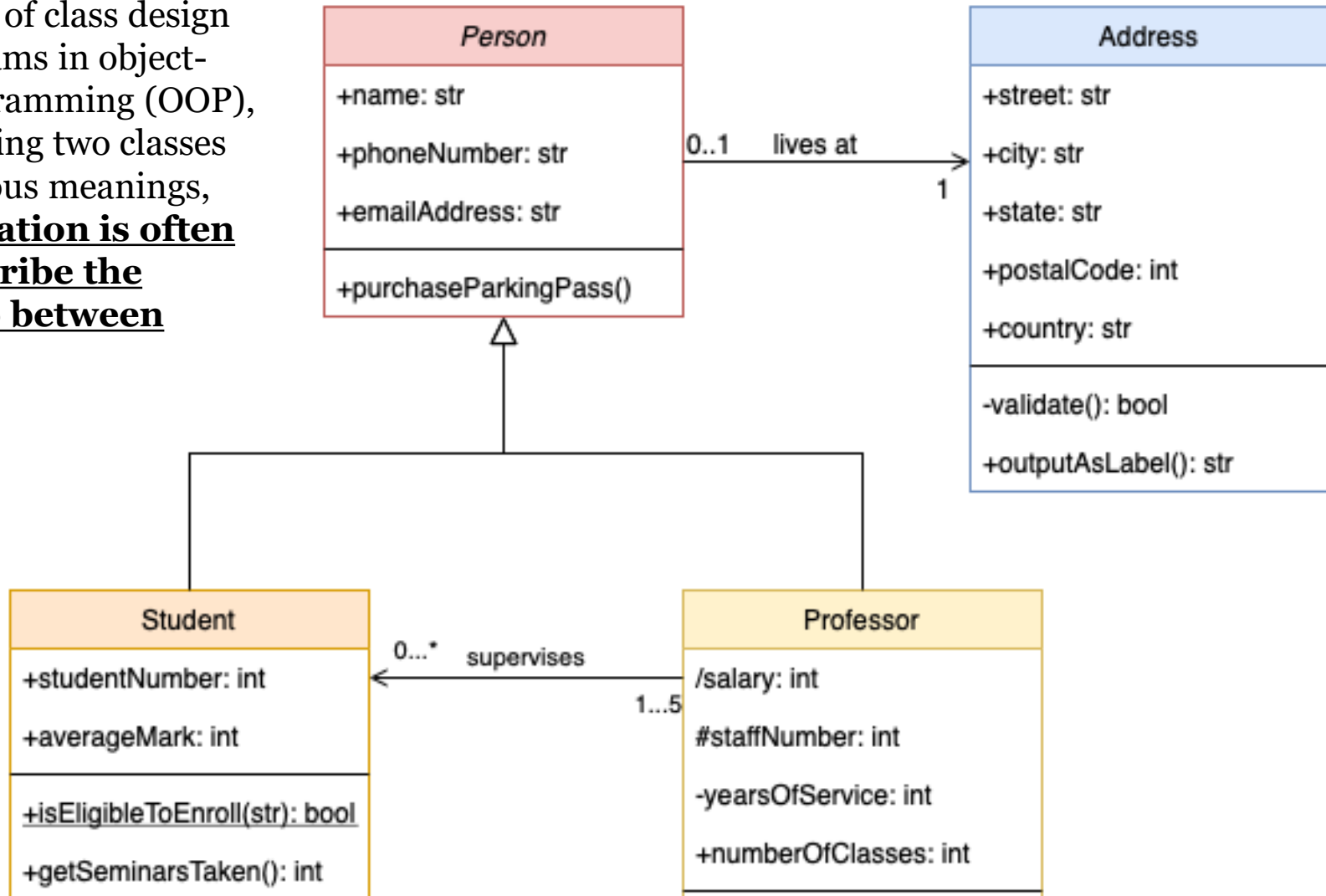
Data Structure in OOP

- ❖ The **blueprint contains the structure that will hold the pieces of data**. A blueprint or class contains properties (*name, age, eye color*), a constructor, and behavior/function.
- ❖ **Object-Oriented Programming (OOP)** and **class diagrams** in the **Unified Modeling Language (UML)** are interrelated. UML is a **visual modeling language** used to **document**, **design**, and **model** software systems, including systems built with the OOP paradigm.
- ❖ One type of UML diagram used to show the static structure of a software system is the **UML class diagram**. In this context, classes in OOP are very important.

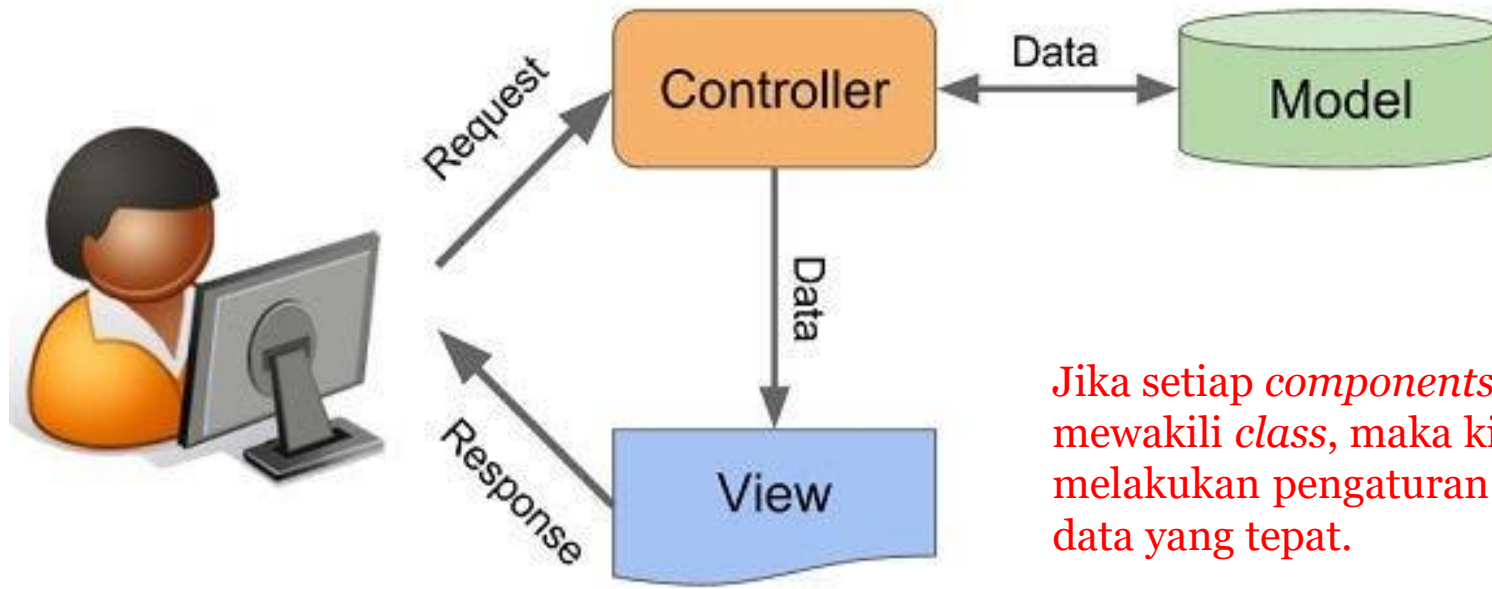


Example Class Diagram

In the context of class design or class diagrams in object-oriented programming (OOP), a line connecting two classes can have various meanings, **and this notation is often used to describe the relationship between classes.**



Perlu untuk diingat pada bagian awal ini, yaitu tujuan utama *data structure* adalah untuk memungkinkan pengaksesan dan manipulasi data dengan efisien.

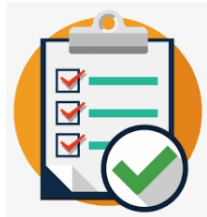


Jika setiap *components* akan mewakili *class*, maka kita harus melakukan pengaturan pengaksesan data yang tepat.

- ❖ Dalam OOP, sebuah *object* akan memiliki **atribut (data)** dan **method/metode (fungsi)** yang bekerja pada data.
- ❖ Paradigma pemrograman OOP ini menggunakan konsep *object*, dimana menggabungkan **DATA** dan **FUNGSI** terkait ke dalam satu entitas yang disebut *object*.
- ❖ Karena *object* dalam OOP sering kali menggunakan *data structure* untuk menyimpan dan mengelola data internal. Contohnya, *object* dapat memiliki *attribute* yang mewakili data, dan *method* yang melakukan operasi pada data tersebut.

Test your understanding

The following are some multiple choice questions to reinforce your understanding of the concepts we've covered so far.



Question #1

Apa yang di maksud dengan **data structure** atau struktur data? Jelaskan hubungan data structure dan object oriented programming (OOP)?

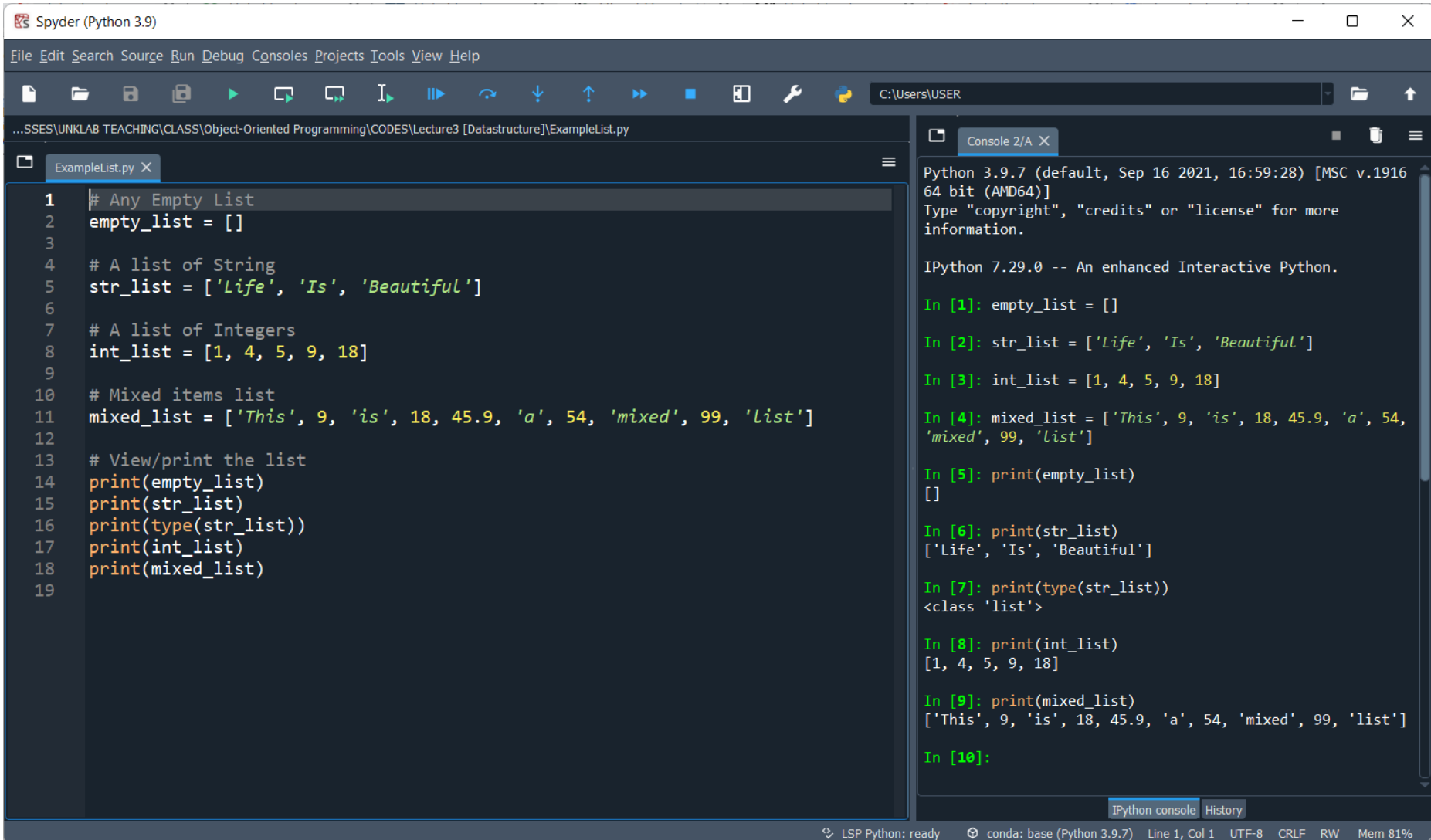
Question #2

Object Oriented Programming (OOP) adalah tentang memperlakukan data sebagai objek (*objects*). Blueprint atau disebut juga Class (*di dalam code program*) akan berisi **properties** (name, age, eye color), **constructor** dan juga **behaviors** (**functions atau methods**). Sebutkan contoh **behaviors** dari class Person?



Exercise *for* Students

[1] Data Structure - List



The image shows the Spyder Python IDE interface. The left pane displays a Python script named `ExampleList.py` with the following code:

```
1 # Any Empty List
2 empty_list = []
3
4 # A list of String
5 str_list = ['Life', 'Is', 'Beautiful']
6
7 # A list of Integers
8 int_list = [1, 4, 5, 9, 18]
9
10 # Mixed items list
11 mixed_list = ['This', 9, 'is', 18, 45.9, 'a', 54, 'mixed', 99, 'list']
12
13 # View/print the list
14 print(empty_list)
15 print(str_list)
16 print(type(str_list))
17 print(int_list)
18 print(mixed_list)
19
```

The right pane shows the IPython console output for the script:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v.1916
64 bit (AMD64)]
Type "copyright", "credits" or "license" for more
information.

IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: empty_list = []

In [2]: str_list = ['Life', 'Is', 'Beautiful']

In [3]: int_list = [1, 4, 5, 9, 18]

In [4]: mixed_list = ['This', 9, 'is', 18, 45.9, 'a', 54,
'mixed', 99, 'list']

In [5]: print(empty_list)
[]

In [6]: print(str_list)
['Life', 'Is', 'Beautiful']

In [7]: print(type(str_list))
<class 'list'>

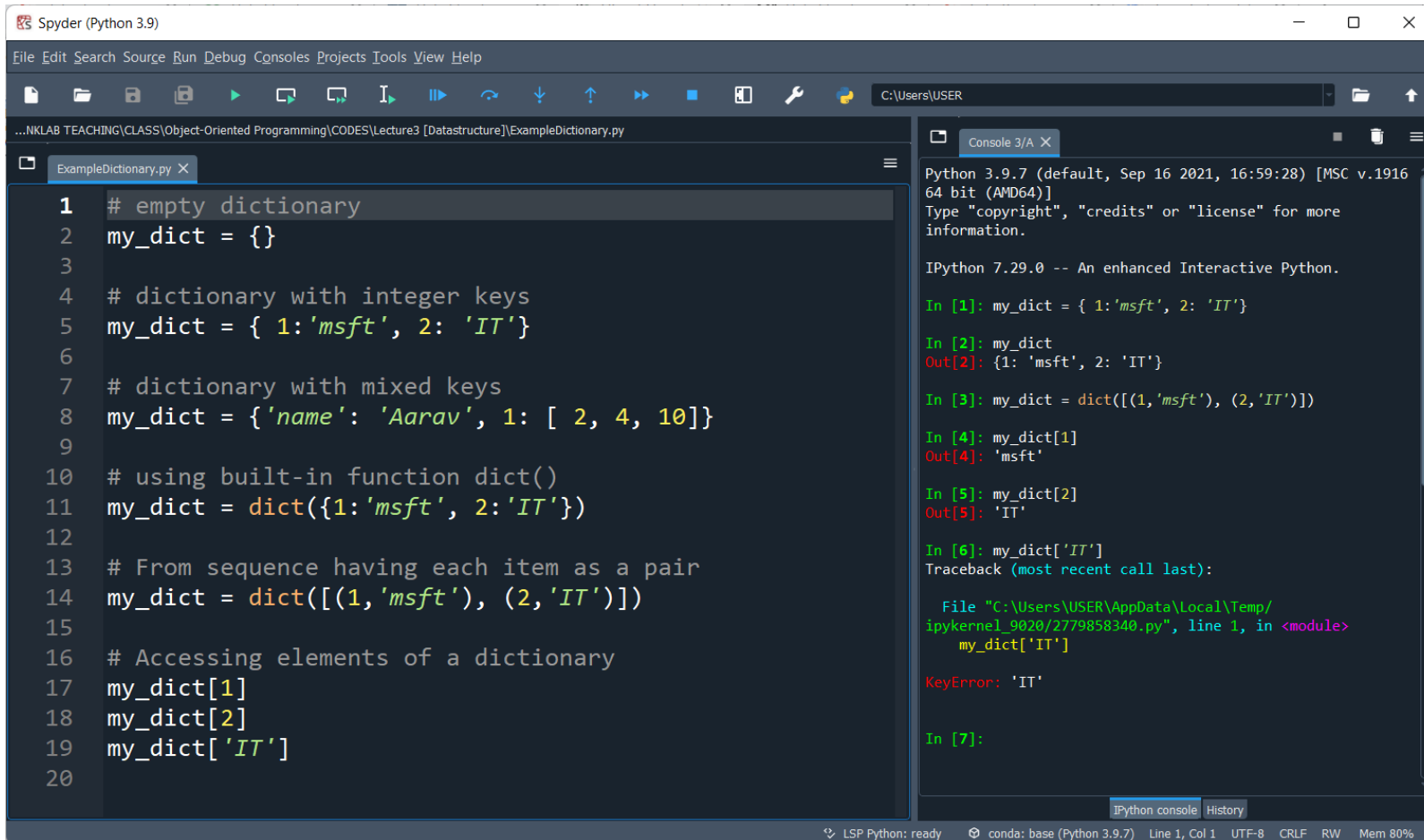
In [8]: print(int_list)
[1, 4, 5, 9, 18]

In [9]: print(mixed_list)
['This', 9, 'is', 18, 45.9, 'a', 54, 'mixed', 99, 'list']

In [10]:
```

The status bar at the bottom indicates: LSP Python: ready, conda: base (Python 3.9.7), Line 1, Col 1, UTF-8, CRLF, RW, Mem 81%.

[2] Data Structure - Dictionary



The screenshot shows the Spyder Python IDE interface. The left pane displays a Python script named `ExampleDictionary.py` with the following code:

```
1 # empty dictionary
2 my_dict = {}
3
4 # dictionary with integer keys
5 my_dict = { 1:'msft', 2: 'IT'}
6
7 # dictionary with mixed keys
8 my_dict = {'name': 'Aarav', 1: [ 2, 4, 10]}
9
10 # using built-in function dict()
11 my_dict = dict({1:'msft', 2:'IT'})
12
13 # From sequence having each item as a pair
14 my_dict = dict([(1,'msft'), (2,'IT')])
15
16 # Accessing elements of a dictionary
17 my_dict[1]
18 my_dict[2]
19 my_dict['IT']
20
```

The right pane shows the IPython console output for the script. The first three lines of code execute successfully, creating the dictionary `my_dict` with integer keys. However, the fourth line, `my_dict[1]`, results in a `KeyError: 'IT'` because the key `1` is not present in the dictionary at that point in the execution (the dictionary still contains the previous state where `1` was a key). The console output shows the following sequence of commands and results:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v.1916
64 bit (AMD64)]
Type "copyright", "credits" or "license" for more
information.

IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: my_dict = { 1:'msft', 2: 'IT'}

In [2]: my_dict
Out[2]: {1: 'msft', 2: 'IT'}

In [3]: my_dict = dict([(1,'msft'), (2,'IT')])

In [4]: my_dict[1]
Out[4]: 'msft'

In [5]: my_dict[2]
Out[5]: 'IT'

In [6]: my_dict['IT']
Traceback (most recent call last):

  File "C:\Users\USER\AppData\Local\Temp/
ipykernel_9020\2779858340.py", line 1, in <module>
    my_dict['IT']
KeyError: 'IT'

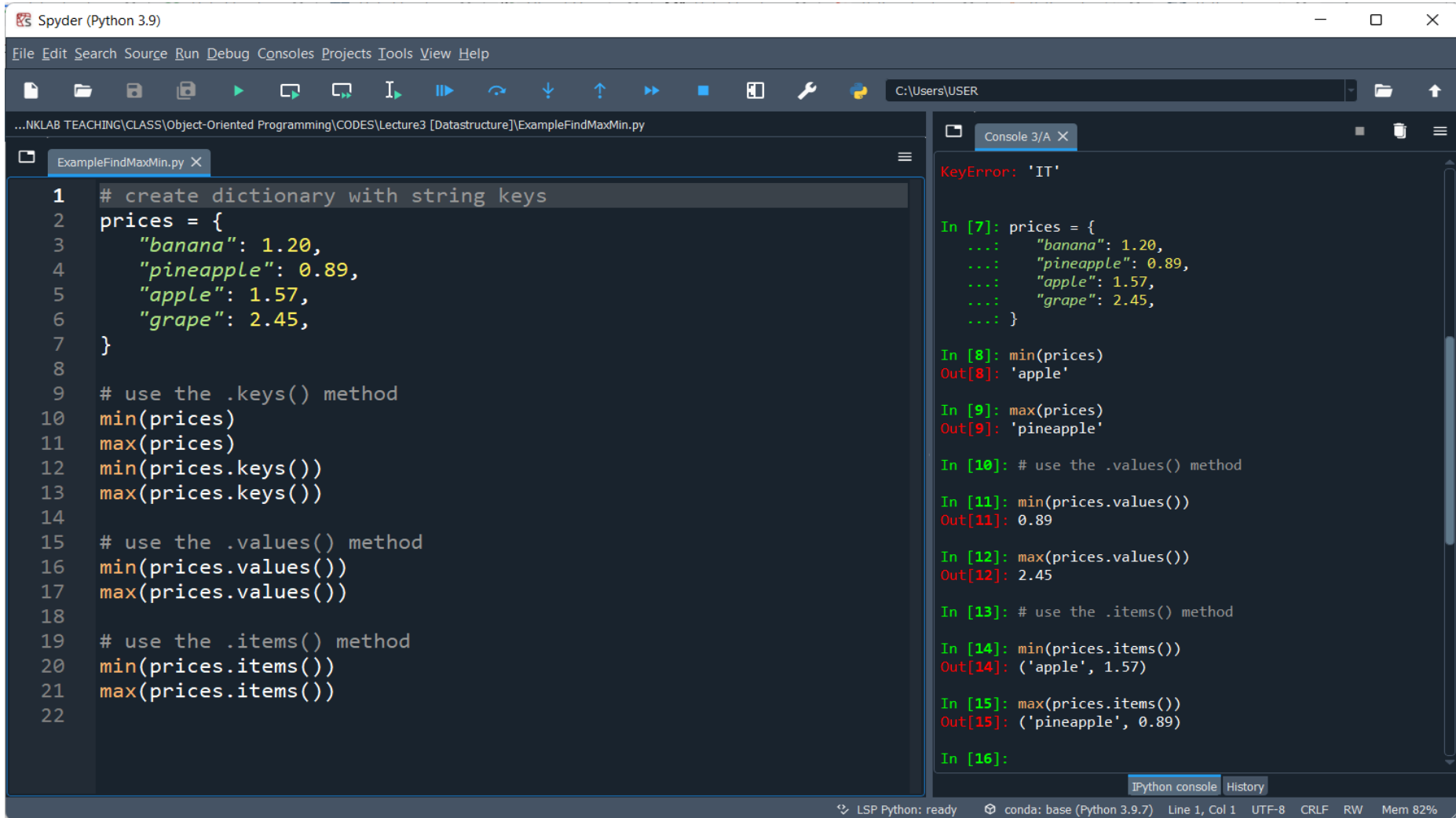
In [7]:
```

The status bar at the bottom indicates the interpreter is ready, the environment is `conda: base (Python 3.9.7)`, and the current file is `Line 1, Col 1`.

Jelaskan mengapa terjadi Error???

[3] Processing Dictionaries With min() & max()

When it comes to processing Python dictionaries with min() and max(), we need to consider that if we use the dictionary directly, then both functions will operate on the keys:



The screenshot shows the Spyder Python IDE interface. The left pane displays a Python script named `ExampleFindMaxMin.py` with the following code:

```
1 # create dictionary with string keys
2 prices = {
3     "banana": 1.20,
4     "pineapple": 0.89,
5     "apple": 1.57,
6     "grape": 2.45,
7 }
8
9 # use the .keys() method
10 min(prices)
11 max(prices)
12 min(prices.keys())
13 max(prices.keys())
14
15 # use the .values() method
16 min(prices.values())
17 max(prices.values())
18
19 # use the .items() method
20 min(prices.items())
21 max(prices.items())
22
```

The right pane shows the IPython console output, which includes a `KeyError: 'IT'` at the top, followed by the execution of the script's code:

```
In [7]: prices = {
...:     "banana": 1.20,
...:     "pineapple": 0.89,
...:     "apple": 1.57,
...:     "grape": 2.45,
...: }

In [8]: min(prices)
Out[8]: 'apple'

In [9]: max(prices)
Out[9]: 'pineapple'

In [10]: # use the .values() method

In [11]: min(prices.values())
Out[11]: 0.89

In [12]: max(prices.values())
Out[12]: 2.45

In [13]: # use the .items() method

In [14]: min(prices.items())
Out[14]: ('apple', 1.57)

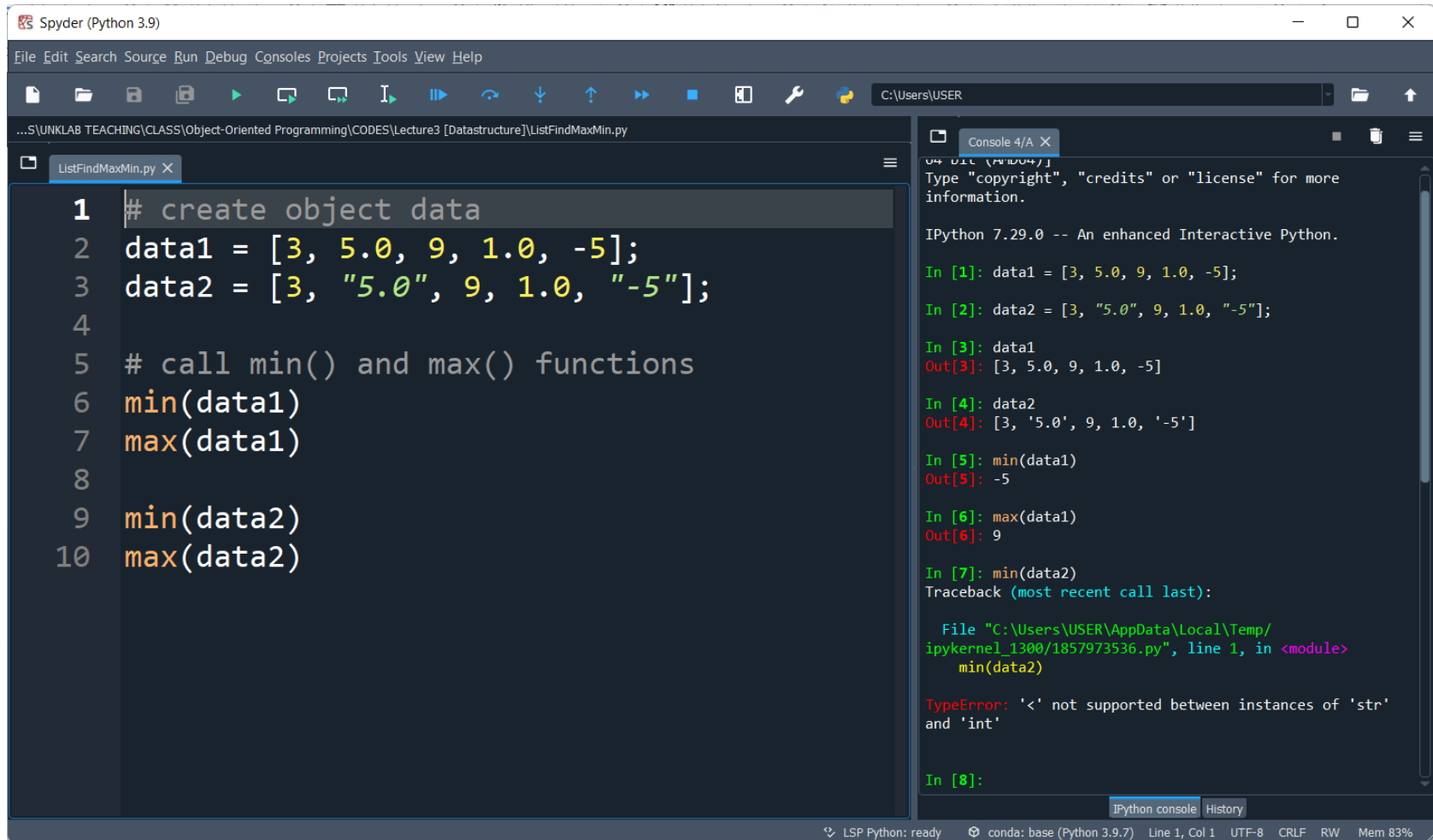
In [15]: max(prices.items())
Out[15]: ('pineapple', 0.89)

In [16]:
```

The status bar at the bottom indicates the environment is `LSP Python: ready`, the conda base (Python 3.9.7) is active, and the current file is `Line 1, Col 1` in `UTF-8` encoding with `CRLF` line endings. The memory usage is `Mem 82%`.

[4] Find smallest & largest values *in* List

Use Python's `min()` and `max()` to find smallest and largest values in List data.



The screenshot shows the Spyder Python IDE interface. The left pane displays a Python script named `ListFindMaxMin.py` with the following code:

```
1 # create object data
2 data1 = [3, 5.0, 9, 1.0, -5];
3 data2 = [3, "5.0", 9, 1.0, "-5"];
4
5 # call min() and max() functions
6 min(data1)
7 max(data1)
8
9 min(data2)
10 max(data2)
```

The right pane shows the IPython console output for the first seven lines of code:

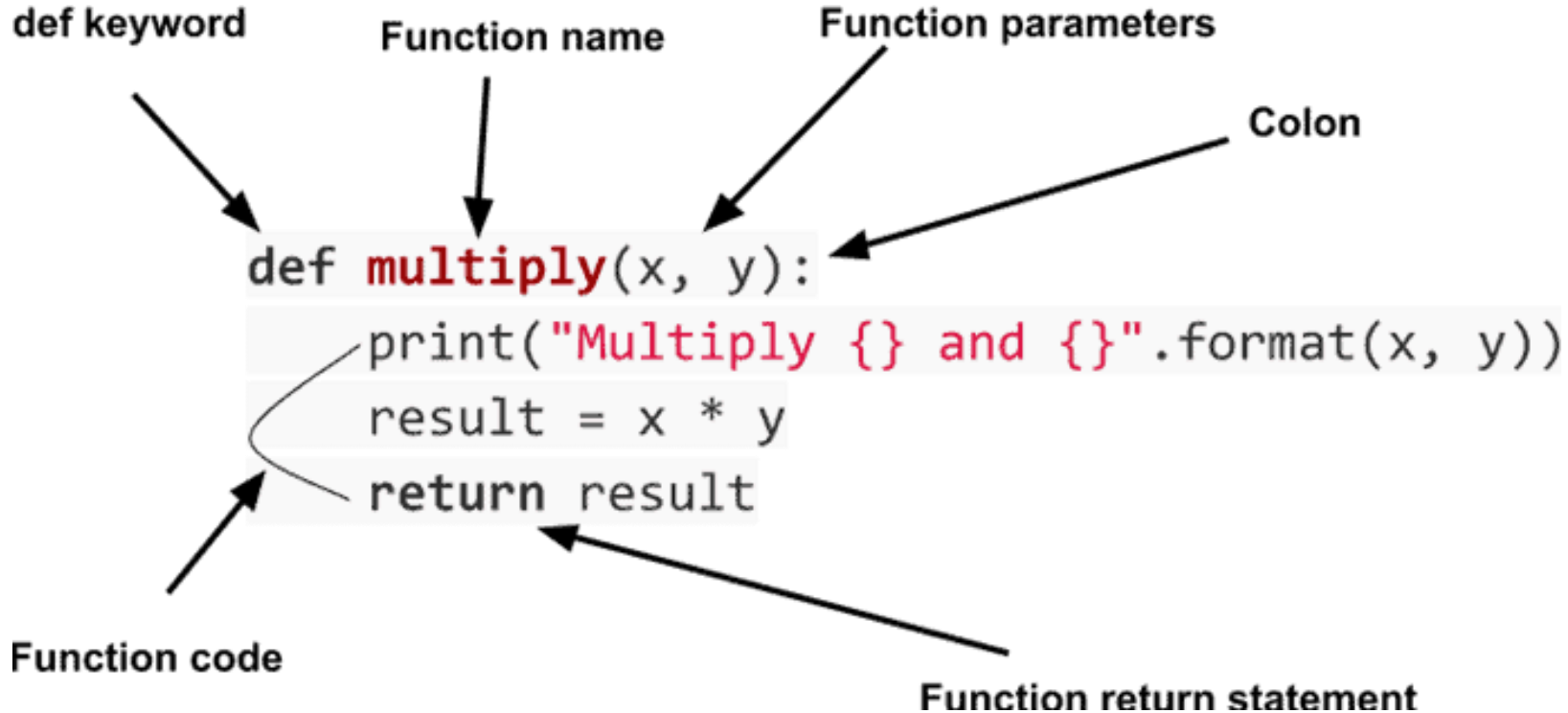
```
In [1]: data1 = [3, 5.0, 9, 1.0, -5];
In [2]: data2 = [3, "5.0", 9, 1.0, "-5"];
In [3]: data1
Out[3]: [3, 5.0, 9, 1.0, -5]
In [4]: data2
Out[4]: [3, '5.0', 9, 1.0, '-5']
In [5]: min(data1)
Out[5]: -5
In [6]: max(data1)
Out[6]: 9
In [7]: min(data2)
Traceback (most recent call last):
  File "C:\Users\USER\AppData\Local\Temp\ipykernel_1300\1857973536.py", line 1, in <module>
    min(data2)
TypeError: '<' not supported between instances of 'str' and 'int'
```

The console shows a `TypeError` for the `min(data2)` call because it attempts to compare a string ('5.0') with an integer (9). The status bar at the bottom indicates the Python version is 3.9.7 and memory usage is 83%.

Jelaskan mengapa terjadi Error???

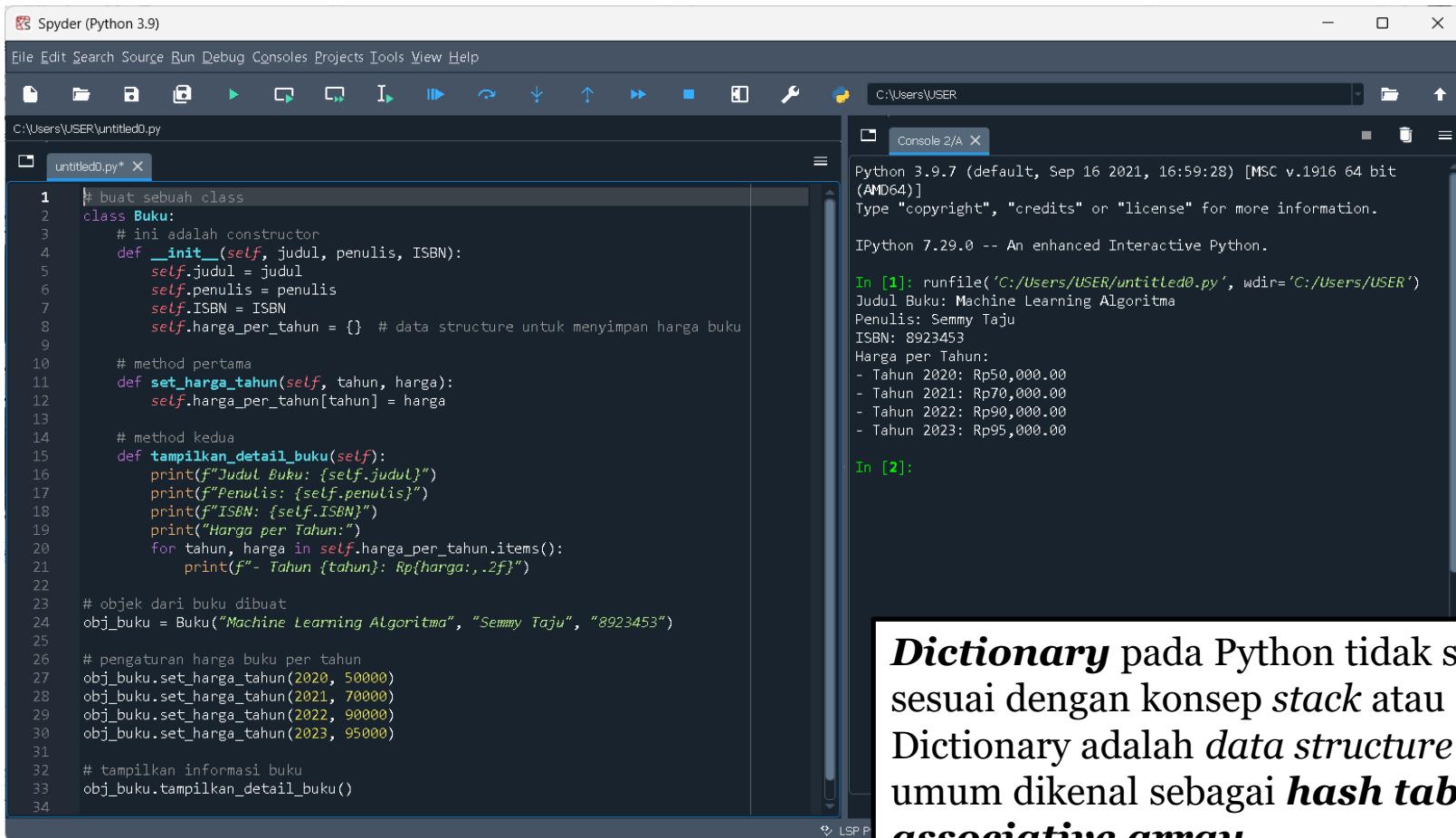
Python Functions

A **Python function** is a block of code which only runs when it is called. You can pass data, known as parameters, into a function. “**def**” keyword pada Python adalah singkatan dari “*define*” yang menunjukkan bahwa kita sedang mendefinisikan sebuah fungsi.



[5] Data Structure in Class

The *data structure* used is dictionary (*harga_per_tahun*). Dictionary is a *data structure* that allows us to store key-value pairs.



The screenshot shows the Spyder Python IDE with a file named `untitled0.py`. The code defines a `Buku` class with an `__init__` method and two other methods: `set_harga_tahun` and `tampilkan_detail_buku`. The `__init__` method initializes attributes `judul`, `penulis`, `ISBN`, and a dictionary `harga_per_tahun`. The `set_harga_tahun` method updates the price for a given year. The `tampilkan_detail_buku` method prints the book details and the price for each year. An object `obj_buku` is created with the title "Machine Learning Algoritma", author "Semmy Taju", and ISBN "8923453". The price for each year (2020 to 2023) is set. Finally, the `tampilkan_detail_buku` method is called on the object.

```
1 # buat sebuah class
2 class Buku:
3     # ini adalah constructor
4     def __init__(self, judul, penulis, ISBN):
5         self.judul = judul
6         self.penulis = penulis
7         self.ISBN = ISBN
8         self.harga_per_tahun = {} # data structure untuk menyimpan harga buku
9
10    # method pertama
11    def set_harga_tahun(self, tahun, harga):
12        self.harga_per_tahun[tahun] = harga
13
14    # method kedua
15    def tampilkan_detail_buku(self):
16        print(f"Judul Buku: {self.judul}")
17        print(f"Penulis: {self.penulis}")
18        print(f"ISBN: {self.ISBN}")
19        print("Harga per Tahun:")
20        for tahun, harga in self.harga_per_tahun.items():
21            print(f"- Tahun {tahun}: Rp{harga:,.2f}")
22
23    # objek dari buku dibuat
24    obj_buku = Buku("Machine Learning Algoritma", "Semmy Taju", "8923453")
25
26    # pengaturan harga buku per tahun
27    obj_buku.set_harga_tahun(2020, 50000)
28    obj_buku.set_harga_tahun(2021, 70000)
29    obj_buku.set_harga_tahun(2022, 90000)
30    obj_buku.set_harga_tahun(2023, 95000)
31
32    # tampilkan informasi buku
33    obj_buku.tampilkan_detail_buku()
34
```

The console output shows the execution of the code:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v.1916 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/USER/untitled0.py', wdir='C:/Users/USER')
Judul Buku: Machine Learning Algoritma
Penulis: Semmy Taju
ISBN: 8923453
Harga per Tahun:
- Tahun 2020: Rp50,000.00
- Tahun 2021: Rp70,000.00
- Tahun 2022: Rp90,000.00
- Tahun 2023: Rp95,000.00

In [2]:
```

Dictionary pada Python tidak sepenuhnya sesuai dengan konsep *stack* atau *queue*. Dictionary adalah *data structure* yang lebih umum dikenal sebagai **hash table** atau **associative array**.

Perhatikan class OOP di atas, jelaskan *structure* dari class OOP tersebut?

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

